

The Collected Mathematical Papers of Arthur Cayley

Volume I

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[Continuation of paper 37: “On the Rotation of a Solid Body round a Fixed Point.”]

Also λ, μ, ν are given by the equations (14) as functions of p, q, r, Ω , i.e. of v, Ω ; so that every thing is prepared for the investigation of the differential equation between V, Ω . To find this we have immediately

$$dv = \frac{1}{2}(Aa dp + Bb dq + Cc dr) = \frac{1}{2}\nabla dt, \quad (19)$$

from the equations (4) and (15). ∇ is of course to be considered as a given function of v . Again,

$$\Omega d\Omega = \frac{1}{2}(\kappa dv + v d\kappa), \quad (20)$$

where

$$d\kappa = 2(\lambda d\lambda + \mu d\mu + \nu d\nu); \quad (21)$$

or from the equations (1), (3), [and (3 bis)],

$$d\kappa = \kappa(\lambda p + \mu q + \nu r) dt. \quad (22)$$

Hence, from (16),

$$2v d\kappa = \kappa\{\Omega(h + \Phi) - \nabla\} dt; \quad (23)$$

$$2(v d\kappa + \kappa dv) = \kappa\Omega(h + \Phi) dt, \quad (24)$$

whence

$$\begin{aligned} d\Omega &= \frac{1}{2}\kappa(h + \Phi) dt \\ &= \frac{1}{2}\frac{\Omega^2 + k^2}{v}(h + \Phi) dt, \end{aligned} \quad (25)$$

and therefore, from (19),

$$\frac{2d\Omega}{\Omega^2 + k^2} = \frac{h + \Phi}{v\nabla} dv, \quad (26)$$

the required differential equation, in which Φ, ∇ are given functions of v , i.e. they are functions of p, q, r by the equations (15), and these quantities are functions of v by (18). The variables in (26) are therefore separated, and we have the integral equation

$$\frac{2}{k} \tan^{-1} \frac{\Omega}{k} = \mathfrak{S} + k \int \frac{h + \Phi}{v\nabla} dv, \quad (27)$$

where $\mathfrak{S}$ is the constant of integration. The equation (19) gives also

$$t - \epsilon = 2 \int \frac{dv}{\nabla}; \quad (28)$$

and thus the solution of the problem is completely effected. The integrals may be taken from any particular value v_0 of v . The variable Ω may be exhibited as the integral of an explicit algebraical function, by recurring to the variable ϕ of the paper quoted.

Thus if

$$\begin{aligned} Ap_0^2 + Bq_0^2 + Cr_0^2 &= h, \\ A^2p_0^2 + B^2q_0^2 + C^2r_0^2 &= k^2, \\ Ap_0a + Bq_0b + Cr_0c &= 2v_0 - k^2; \end{aligned}$$

then the values of p, q, r are respectively

$$\sqrt{\{p_0^2 + \frac{1}{A}(C - B)\phi\}}, \quad \sqrt{\{q_0^2 + \frac{1}{B}(A - C)\phi\}}, \quad \sqrt{\{r_0^2 + \frac{1}{C}(B - A)\phi\}},$$

where

$$dt = \frac{1}{2} \frac{d\phi}{pqr} = \frac{2 dv}{\nabla}, \quad \text{or} \quad \frac{dv}{\nabla} = \frac{1}{4} \frac{d\phi}{pqr};$$

and then

$$\frac{2}{k} \tan^{-1} \frac{\Omega}{k} = 2\mathfrak{S} + k \int \frac{(h + ap + bq + cr) d\phi}{(k^2 + Apa + Bqb + Crc) pqr}.$$

in which form it is exactly analogous to the equation there obtained, p. 230, [6, p. 34]

$$4 \tan^{-1} \frac{\Omega}{k} \Big|_{v_0}^v = \int \frac{(h + Cr) d\phi}{(k^2 + Cr) pqr}.$$

On the Variation of the Constants, when the body is acted upon by Forces.

The dynamical equations of a problem being expressed in the form

$$\begin{aligned} \frac{d}{dt} \frac{dT}{d\lambda'} &= \frac{dT}{d\lambda} - \frac{dV}{d\lambda}, \\ \frac{d}{dt} \frac{dT}{d\mu'} &= \frac{dT}{d\mu} - \frac{dV}{d\mu}, \\ \frac{d}{dt} \frac{dT}{d\nu'} &= \frac{dT}{d\nu} - \frac{dV}{d\nu}, \end{aligned}$$

suppose the equations obtained from these by neglecting the function V , are integrated; each of the six integrals may be expressed in the form

$$a = f(\lambda, \mu, \nu, \lambda', \mu', \nu', t),$$

where a denotes any one of the arbitrary constants. Assume

$$\frac{dT}{d\lambda'} = u, \quad \frac{dT}{d\mu'} = v, \quad \frac{dT}{d\nu'} = w;$$

then λ', μ', ν' may be expressed in terms of $\lambda, \mu, \nu, u, v, w$, and the integrals may be reduced to the form

$$a = F(\lambda, \mu, \nu, u, v, w, t).$$

These equations may be considered as the integrals of the proposed system, taking into account the terms involving V , provided the constants [say a, b, c, d, e, f] be sup

posed to become variable. We have, in this case, by Lagrange's theory of the variation of the arbitrary constants, the formulae

$$\frac{da}{dt} = (a, b) \frac{dV}{db} + (a, c) \frac{dV}{dc} + (a, d) \frac{dV}{dd} + (a, e) \frac{dV}{de} + (a, f) \frac{dV}{df};$$

where

$$(a, b) = \left(\frac{da}{du} \frac{db}{d\lambda} - \frac{da}{d\lambda} \frac{db}{du} \right) + \left(\frac{da}{dv} \frac{db}{d\mu} - \frac{da}{d\mu} \frac{db}{dv} \right) + \left(\frac{da}{dw} \frac{db}{d\nu} - \frac{da}{d\nu} \frac{db}{dw} \right),$$

and in which V is supposed to be expressed as a function of a, b, c, d, e, f, t .

Thus the solution of the problem requires the calculation of thirty coefficients (a, b) , or rather of fifteen only, since evidently $(a, b) = -(b, a)$. It is known that these coefficients are functions of a, b, c, d, e, f , without t ; so that, in calculating them, any assumed arbitrary value, e.g. $t = 0$, may be given to the time.

In practice, it often happens that one of the arbitrary constants, e.g. a , may be expressed in the form

$$a = F(\lambda, \mu, \nu, u, v, w, t, b, c, d, e, f),$$

where b, c, d, e, f are given functions of $\lambda, \mu, \nu, u, v, w, t$. In this case, it is easily seen that we may write

$$(a, b) = \{(a, b)\} + (c, b) \frac{da}{dc} + (d, b) \frac{da}{dd} + (e, b) \frac{da}{de} + (f, b) \frac{da}{df},$$

where, in the calculation of $\{(a, b)\}$, the differentiations upon a are performed without taking into account the variability of $b, c, \dots$

In the particular problem in question, the following are the values of the new variables u, v, w (*Math. Journal*, memoir already quoted, [6]),

$$u = \frac{2}{\kappa} (Ap - \nu Bq + \mu Cr), \tag{29}$$

$$v = \frac{2}{\kappa} (\nu Ap + Bq - \lambda Cr),$$

$$w = \frac{2}{\kappa} (-\mu Ap + \lambda Bq + Cr);$$

equations which may also be expressed in the form

$$2Ap = (1 + \lambda^2)u + (\lambda\mu + \nu)v + (\nu\lambda - \mu)w, \tag{30}$$

$$2Bq = (\lambda\mu - \nu)u + (1 + \mu^2)v + (\mu\nu + \lambda)w,$$

$$2Cr = (\nu\lambda + \mu)u + (\mu\nu - \lambda)v + (1 + \nu^2)w,$$

or putting for shortness

$$\lambda u + \mu v + \nu w = \varpi, \tag{31}$$

these become

$$\begin{aligned} 2Ap &= \lambda\varpi + u + \nu v - \mu w, \\ 2Bq &= \mu\varpi - \nu u + v + \lambda w, \\ 2Cr &= \nu\varpi + \mu u - \lambda v + w, \end{aligned} \tag{32}$$

whence also

$$2\Omega = \kappa\varpi. \tag{33}$$

Substituting the values of Ap , Bq , Cr , given by (30) in the equations (6), we deduce

$$\begin{aligned} 2\mathfrak{a} &= \lambda\varpi + u - \nu v + \mu w, \\ 2\mathfrak{b} &= \mu\varpi + \nu u + v - \lambda w, \\ 2\mathfrak{c} &= \nu\varpi - \mu u + \lambda v + w, \end{aligned} \tag{34}$$

whence also

$$2(\mathfrak{a}\lambda + \mathfrak{b}\mu + \mathfrak{c}\nu) = \kappa\varpi, \tag{35}$$

which in fact follows from (33) and (17). And likewise the inverse system,

$$\begin{aligned} u &= \frac{2}{\kappa}(\mathfrak{a} + \nu\mathfrak{b} - \mu\mathfrak{c}), \\ v &= \frac{2}{\kappa}(-\nu\mathfrak{a} + \mathfrak{b} + \lambda\mathfrak{c}), \\ w &= \frac{2}{\kappa}(\mu\mathfrak{a} - \lambda\mathfrak{b} + \mathfrak{c}). \end{aligned} \tag{36}$$

It is easy to deduce

$$k^2 = \frac{1}{4}\kappa[u^2 + v^2 + w^2 + \varpi^2], \tag{37}$$

$$v = \frac{1}{4}[(u^2 + v^2 + w^2) + (1 + \kappa)\varpi^2]. \tag{38}$$

Again, from the equations (10 bis),

$$\begin{aligned} \kappa(\mathfrak{b}Cr - \mathfrak{c}Bq) &= -2\lambda(\mathfrak{a}^2 + \mathfrak{b}^2 + \mathfrak{c}^2) + 2\mathfrak{a}(\lambda\mathfrak{a} + \mu\mathfrak{b} + \nu\mathfrak{c}) + 2(\mathfrak{b}\nu - \mathfrak{c}\mu)\Omega \\ &= -2\lambda k^2 + 2(\mathfrak{a} + \mathfrak{b}\nu - \mathfrak{c}\mu)\Omega \\ &= -2\lambda k^2 + \kappa u\Omega; \end{aligned}$$

and, forming also the similar expressions for $\kappa(\mathfrak{c}Ap - \mathfrak{a}Cr)$, and $\kappa(\mathfrak{a}Bq - \mathfrak{b}Ap)$, we thus obtain,

$$\begin{aligned} \Omega u - \frac{2}{\kappa}k^2\lambda &= \mathfrak{b}Cr - \mathfrak{c}Bq, \\ \Omega v - \frac{2}{\kappa}k^2\mu &= \mathfrak{c}Ap - \mathfrak{a}Cr, \\ \Omega w - \frac{2}{\kappa}k^2\nu &= \mathfrak{a}Bq - \mathfrak{b}Ap; \end{aligned} \tag{39}$$

to which many others might probably be joined.

The constants of the problem are $\mathfrak{a}$, $\mathfrak{b}$, $\mathfrak{c}$, h , ϵ , $\mathfrak{S}$. Of these $\mathfrak{a}$, $\mathfrak{b}$, $\mathfrak{c}$ are given as functions of λ , μ , ν , u , v , w , by the equations (34); in which ϖ is to be considered as

standing for $\lambda u + \mu v + \nu w$. (These determine k^2 , which is however given immediately by (37).) As for h , we have

$$h = \frac{1}{A}(Ap)^2 + \frac{1}{B}(Bq)^2 + \frac{1}{C}(Cr)^2,$$

where Ap, Bq, Cr are given as functions of $\lambda, \mu, \nu, u, v, w$ by (32), in which also ϖ stands for $\lambda u + \mu v + \nu w$. Again,

$$\epsilon = t - \frac{1}{2} \int \frac{dv}{\nabla},$$

$$\mathfrak{S} = \frac{2}{k} \tan^{-1} \frac{\kappa \varpi}{2k} - k \int \frac{(h + \Phi) dv}{v \nabla};$$

in each of which ∇, Φ are functions of v , and of $\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, h$, partly as entering explicitly into these functions, partly as contained implicitly in p, q, r , which enter into ∇, Φ , and into functions of v, κ given by (18). After the integration v is to be considered a function of $\lambda, \mu, \nu, u, v, w$ given by (38). Both of the integrals may be supposed taken from a certain value v_0 of v , which may be considered as an absolutely invariable arbitrary constant, since without it we have the right number, six, of arbitrary constants.

First to find $(\mathfrak{a}, \mathfrak{b})$, $(\mathfrak{b}, \mathfrak{c})$, and $(\mathfrak{c}, \mathfrak{a})$. From (34) we have

$$\begin{aligned} (\mathfrak{a}, \mathfrak{b}) &= \frac{1}{4} \{ (1 + \lambda^2)(\mu u - w) - (\lambda u + \varpi)(\nu + \lambda) \\ &\quad + (\lambda \mu - \nu)(\mu v + \varpi) - (\lambda w + u)(1 + \mu^2) \\ &\quad + (\nu \lambda + \mu)(u + \mu w) - (\lambda w - v)(\mu v - \lambda) \} \\ &= \frac{1}{4}(\mu w - \lambda v - u - \nu \varpi) = -\frac{1}{2} \cdot 2\mathfrak{c} = -\mathfrak{c}; \end{aligned}$$

whence the system

$$(\mathfrak{b}, \mathfrak{c}) = -\mathfrak{a}, \quad (\mathfrak{c}, \mathfrak{a}) = -\mathfrak{b}, \quad (\mathfrak{a}, \mathfrak{b}) = -\mathfrak{c}. \quad (40)$$

Also we may add

$$(k^2, \mathfrak{a}) = \frac{\mathfrak{a}}{k}(\mathfrak{a}, \mathfrak{a}) + \frac{\mathfrak{b}}{k}(\mathfrak{b}, \mathfrak{a}) + \frac{\mathfrak{c}}{k}(\mathfrak{c}, \mathfrak{a}) = 0,$$

or

$$(k, \mathfrak{a}) = 0, \quad (k, \mathfrak{b}) = 0, \quad (k, \mathfrak{c}) = 0, \quad (41)$$

which will be useful in calculating some of the following coefficients.

Proceeding to calculate $(\mathfrak{a}, h)$, $(\mathfrak{b}, h)$, $(\mathfrak{c}, h)$. It is seen immediately that

$$(\mathfrak{a}, h) = 2\{p(\mathfrak{a}, Ap) + q(\mathfrak{a}, Bq) + r(\mathfrak{a}, Cr)\},$$

where Ap, Bq, Cr , are given by the equations (32), so that

$$\begin{aligned} (\mathfrak{a}, Ap) &= \frac{1}{4} [(1 + \lambda^2)(\lambda u + \varpi) - (1 + \lambda^2)(\lambda u + \varpi) \\ &\quad + (\lambda \mu - \nu)(\lambda v - w) - (\lambda \mu + \nu)(\lambda v + w) \\ &\quad + (\nu \lambda + \mu)(\nu + \lambda w) - (\nu \lambda - \mu)(-v + \lambda w)] \end{aligned}$$

i.e.

$$(\mathfrak{a}, Ap) = 0. \quad (42)$$

Similarly

$$\begin{aligned} (\mathfrak{a}, Bq) = \frac{1}{4} [& (1 + \lambda^2)(\mu u + w) - (\lambda u + \varpi)(\lambda \mu - \nu) \\ & + (\lambda \mu - \nu)(\mu v + \varpi) - (\lambda v + w)(1 + \mu^2) \\ & + (\nu \lambda + \mu)(\mu w - v) - (-v + \lambda w)(\mu v + \lambda)] \end{aligned}$$

i.e.

$$(\mathfrak{a}, Bq) = 0, \quad (43)$$

and similarly

$$(\mathfrak{a}, Cr) = 0, \quad (44)$$

whence $(\mathfrak{a}, h) = 0$, and therefore $(\mathfrak{b}, h) = 0$, $(\mathfrak{c}, h) = 0$; also

$$(k, h) = 0. \quad (45)$$

Next we have to determine $(\mathfrak{a}, e)$, $(\mathfrak{b}, e)$, $(\mathfrak{c}, e)$. Here e being a function of $u, v, w, \lambda, \mu, \nu, \mathfrak{a}, \mathfrak{b}, \mathfrak{c}, h$, we must write

$$(\mathfrak{a}, e) = \{(\mathfrak{a}, e)\} + (\mathfrak{a}, \mathfrak{b}) \frac{de}{d\mathfrak{b}} + (\mathfrak{a}, \mathfrak{c}) \frac{de}{d\mathfrak{c}} + (\mathfrak{a}, h) \frac{de}{dh},$$

i.e.

$$(\mathfrak{a}, e) = \{(\mathfrak{a}, e)\} + \mathfrak{b} \frac{de}{d\mathfrak{c}} - \mathfrak{c} \frac{de}{d\mathfrak{b}}.$$

But

$$e = t - \frac{1}{2} \int \frac{dv}{\nabla}; \quad \text{and thence} \quad \{(\mathfrak{a}, e)\} = -\frac{2}{\nabla}(\mathfrak{a}, v),$$

and v is given immediately as a function of $\lambda, \mu, \nu, u, v, w$, by the equation (38). Hence

$$\begin{aligned} (\mathfrak{a}, v) &= \frac{1}{4} [(1 + \lambda^2) \{ (1 + \kappa) u \varpi + \lambda \varpi^2 \} - (\lambda u + \varpi) [u + \lambda (1 + \kappa) \varpi] \\ &\quad + (\lambda \mu - \nu) \{ (1 + \kappa) v \varpi + \mu \varpi^2 \} - (\lambda v + w) [v + \mu (1 + \kappa) \varpi] \\ &\quad + (\nu \lambda + \mu) \{ (1 + \kappa) w \varpi + \nu \varpi^2 \} - (-v + \lambda w) [w + \nu (1 + \kappa) \varpi]] \\ &= \frac{1}{4} \{ (1 + \kappa) \varpi u - \lambda (1 + \kappa) \varpi^2 + \lambda \kappa \varpi^2 - \lambda (u^2 + v^2 + w^2) - \varpi u \} \\ &= \frac{1}{4} \{ \kappa u \varpi - \lambda \varpi^2 - \lambda (u^2 + v^2 + w^2) \} \\ &= \frac{1}{4} \kappa u \varpi - \frac{1}{\kappa} k^2 \left(\lambda \frac{\kappa \varpi}{2} \right) \frac{1}{2} \kappa \quad \{ \text{by (37) and (33)} \}, \\ &= \frac{1}{2} (\mathfrak{b} Cr - \mathfrak{c} Bq); \end{aligned} \quad (46)$$

whence

$$\{(\mathfrak{a}, e)\} = -\frac{1}{\nabla} (\mathfrak{b} Cr - \mathfrak{c} Bq),$$

$$(\mathfrak{a}, e) = -\frac{1}{\nabla} (\mathfrak{b} Cr - \mathfrak{c} Bq) + \mathfrak{b} \frac{de}{d\mathfrak{c}} - \mathfrak{c} \frac{de}{d\mathfrak{b}}.$$

The terms $\mathfrak{b} \frac{de}{d\mathfrak{c}} - \mathfrak{c} \frac{de}{d\mathfrak{b}}$ are evidently of the form $F(v) - F(v_0)$.

If therefore we suppose $v = v_0$, we have

$$(\mathfrak{a}, e) = -\frac{1}{\nabla_0} (\mathfrak{b} Cr_0 - \mathfrak{c} Bq_0), \quad (47)$$

if p_0, q_0, r_0, ∇_0 refer to the value v_0 of v , i.e. if

$$\begin{aligned} Ap_0^2 + Bq_0^2 + Cr_0^2 &= h, \\ A^2p_0^2 + B^2q_0^2 + C^2r_0^2 &= k^2, \\ Ap_0\mathfrak{a} + Bq_0\mathfrak{b} + Cr_0\mathfrak{c} &= 2v_0 - k^2. \end{aligned} \tag{48}$$

[This implies evidently

$$\mathfrak{b}\frac{de}{dc} - \mathfrak{c}\frac{de}{db} = \frac{1}{\nabla}(\mathfrak{b}Cr - \mathfrak{c}Bq) - \frac{1}{\nabla_0}(\mathfrak{b}Cr_0 - \mathfrak{c}Bq_0),$$

an equation which it is interesting to verify. In fact, from the value of e

$$\mathfrak{b}\frac{de}{dc} - \mathfrak{c}\frac{de}{db} = -2 \int dv \left(\mathfrak{b}\frac{d}{dc} - \mathfrak{c}\frac{d}{db} \right) \frac{1}{\nabla} = -2 \int dv \frac{1}{\nabla^2} \left(\mathfrak{b}\frac{d\nabla}{dc} - \mathfrak{c}\frac{d\nabla}{db} \right);$$

or we have to show that

$$\frac{d}{dv} \frac{1}{\nabla} (\mathfrak{b}Cr - \mathfrak{c}Bq) = \frac{2}{\nabla^2} \left(\mathfrak{b}\frac{d\nabla}{dc} - \mathfrak{c}\frac{d\nabla}{db} \right) = \frac{2}{\nabla^2} \delta\nabla;$$

if for shortness

$$\delta = \mathfrak{b}\frac{d}{dc} - \mathfrak{c}\frac{d}{db}.$$

Now ∇ containing $\mathfrak{a}, \mathfrak{b}, \mathfrak{c}$ explicitly, and also as involved in p, q, r , we have

$$\delta\nabla = \mathfrak{b}pq(A - B) - \mathfrak{c}rp(C - A) + \frac{d\nabla}{dp}\delta p + \frac{d\nabla}{dq}\delta q + \frac{d\nabla}{dr}\delta r = \mathfrak{b}pq(A - B) - \mathfrak{c}rp(C - A) + \delta'\nabla$$

suppose. The equation to be verified becomes

$$\nabla \left(\mathfrak{b}C\frac{dr}{dv} - \mathfrak{c}B\frac{dq}{dv} \right) - (\mathfrak{b}Cr - \mathfrak{c}Bq)\frac{d\nabla}{dv} = 2\{\mathfrak{b}pq(A - B) - \mathfrak{c}rp(C - A) + \delta'\nabla\}.$$

Now, observing that $\delta k = 0$, we have

$$\begin{aligned} Ap\delta p + Bq\delta q + Cr\delta r &= 0, \\ A^2p\delta p + B^2q\delta q + C^2r\delta r &= 0, \\ A\mathfrak{a}\delta p + B\mathfrak{b}\delta q + C\mathfrak{c}\delta r &= -(\mathfrak{b}Cr - \mathfrak{c}Bq). \end{aligned}$$

Also,

$$\begin{aligned} Ap\frac{dp}{dv} + Bq\frac{dq}{dv} + Cr\frac{dr}{dv} &= 0, \\ A^2p\frac{dp}{dv} + B^2q\frac{dq}{dv} + C^2r\frac{dr}{dv} &= 0, \\ A\mathfrak{a}\frac{dp}{dv} + B\mathfrak{b}\frac{dq}{dv} + C\mathfrak{c}\frac{dr}{dv} &= 2, \end{aligned}$$

whence evidently

$$\frac{dp}{dv} = \frac{-2}{\mathfrak{b}Cr - \mathfrak{c}Bq} \delta p, \quad \frac{dq}{dv} = \frac{-2}{\mathfrak{b}Cr - \mathfrak{c}Bq} \delta q, \quad \frac{dr}{dv} = \frac{-2}{\mathfrak{b}Cr - \mathfrak{c}Bq} \delta r;$$

or

$$\frac{d\nabla}{dv} = \frac{-2}{\mathfrak{b}Cr - \mathfrak{c}Bq} \delta' \nabla;$$

or the equation to be verified is simply

$$\nabla \left(\mathfrak{b}C \frac{dr}{dv} - \mathfrak{c}B \frac{dq}{dv} \right) = 2 \{ \mathfrak{b}pq(A - B) - \mathfrak{c}rp(C - A) \},$$

which follows immediately from the three equations just given for the determination of $\frac{dp}{dv}$, $\frac{dq}{dv}$, $\frac{dr}{dv}$.]

From these values also

$$(k, e) = 0. \quad (49)$$

Next, to calculate (h, e) ,

$$(h, e) = \{(h, e)\} + (h, \mathfrak{a}) \frac{de}{d\mathfrak{a}} + (h, \mathfrak{b}) \frac{de}{d\mathfrak{b}} + (h, \mathfrak{c}) \frac{de}{d\mathfrak{c}};$$

but the three last terms being evidently such as to vanish for $v = v_0$, we may neglect them, and consider (h, e) as the value which $\{(h, e)\}$ assumes for this value of v .

Now

$$\frac{1}{2} \{(h, e)\} = 2p\{(Ap, e)\} + 2q\{(Bq, e)\} + 2r\{(Cr, e)\},$$

where

$$\{(Ap, e)\} = -\frac{2}{\nabla} (Ap, v),$$

and

$$\begin{aligned} (Ap, v) &= \frac{1}{4} \left[(1 + \lambda^2) \{ (1 + \kappa) u \varpi + \lambda \varpi^2 \} - (\lambda u + \varpi) [u + \lambda(1 + \kappa) \varpi] \right. \\ &\quad + (\lambda \mu + \nu) \{ (1 + \kappa) v \varpi + \mu \varpi^2 \} - (\lambda v + \varpi) [v + \mu(1 + \kappa) \varpi] \\ &\quad \left. + (\nu \lambda - \mu) \{ (1 + \kappa) w \varpi + \nu \varpi^2 \} - (v + \lambda w) [w + \nu(1 + \kappa) \varpi] \right] \\ &= \frac{1}{4} \{ (1 + \kappa) \varpi u + \lambda \kappa \varpi^2 - \lambda(1 + \kappa) \varpi^2 - \lambda(u^2 + v^2 + w^2) - \varpi u \} = (\mathfrak{a}, v) \\ &= \frac{1}{2} (\mathfrak{b}Cr - \mathfrak{c}Bq), \end{aligned} \quad (50)$$

whence

$$\{(Ap, e)\} = -\frac{1}{\nabla} (\mathfrak{b}Cr - \mathfrak{c}Bq), \quad (51)$$

and therefore

$$\begin{aligned} \{(Bq, e)\} &= -\frac{1}{\nabla} (\mathfrak{c}Ap - \mathfrak{a}Cr), \\ \{(Cr, e)\} &= -\frac{1}{\nabla} (\mathfrak{a}Bq - \mathfrak{b}Ap), \end{aligned}$$

whence

$$\{(h, e)\} = -2,$$

and therefore

$$(h, e) = -2. \quad (52)$$

Next, to find $(\mathfrak{a}, \mathfrak{S}), (\mathfrak{b}, \mathfrak{S}), (\mathfrak{c}, \mathfrak{S}), (h, \mathfrak{S}),$

$$\mathfrak{S} = \frac{2}{k} \tan^{-1} \frac{\kappa \varpi}{2k} - k \int \frac{(h + \Phi) dv}{v \nabla},$$

$$= \mathfrak{S}' + \mathfrak{S}'' \text{ suppose,}$$

$$(\mathfrak{a}, \mathfrak{S}) = (\mathfrak{a}, \mathfrak{S}') + (\mathfrak{a}, \mathfrak{S}''),$$

$$(\mathfrak{a}, \mathfrak{S}') = \frac{k}{\kappa v} (\mathfrak{a}, \kappa \varpi) + (\mathfrak{a}, k) \frac{d\mathfrak{S}'}{dk},$$

[observing $\kappa^2 \varpi^2 + 4k^2 = 4(\Omega^2 + k^2) = 4\kappa v$],

$$= \frac{k}{\kappa v} (\mathfrak{a}, \kappa \varpi);$$

where

$$\begin{aligned} (\mathfrak{a}, \kappa \varpi) &= \frac{1}{4} \{ (1 + \lambda^2)(\kappa u + 2\lambda \varpi) - (\lambda u + \varpi) \kappa \lambda \\ &\quad + (\lambda \mu - \nu)(\kappa v + 2\mu \varpi) - (\lambda v + \mu) \kappa \mu \\ &\quad + (\nu \lambda + \mu)(\kappa w + 2\nu \varpi) - (-v + \lambda w) \kappa \nu \} \\ &= \frac{1}{4} \kappa (u + \lambda \varpi) = Ap - \nu \frac{\Omega}{\kappa} + \mu Cr \frac{1}{\kappa} + \lambda \Omega = \frac{1}{2} (\mathfrak{a} + Ap) \kappa, \end{aligned} \quad (53)$$

by equations (29), (33), and (10); that is

$$(\mathfrak{a}, \mathfrak{S}') = \frac{k}{2v} (\mathfrak{a} + Ap).$$

Also

$$\begin{aligned} (\mathfrak{a}, \mathfrak{S}'') &= -k \frac{h+\Phi}{v \nabla} (\mathfrak{a}, v) + (\mathfrak{a}, \mathfrak{b}) \frac{d\mathfrak{S}''}{d\mathfrak{b}} + \&c. \\ &= -\frac{1}{2} k \frac{h+\Phi}{v \nabla} (\mathfrak{b} Cr - \mathfrak{c} Bq) + Fv - Fv_0, \end{aligned}$$

whence

$$(\mathfrak{a}, \mathfrak{S}) = \frac{k}{2v} [\mathfrak{a} + Ap - \frac{h+\Phi}{\nabla} (\mathfrak{b} Cr - \mathfrak{c} Bq)] + Fv - Fv_0,$$

or putting $v = v_0$

$$(\mathfrak{a}, \mathfrak{S}) = \frac{k}{2v_0} \left[\mathfrak{a} + Ap_0 - \frac{h+\Phi_0}{\nabla_0} (\mathfrak{b} Cr_0 - \mathfrak{c} Bq_0) \right], \quad (54)$$

and therefore also

$$(\mathfrak{b}, \mathfrak{S}) = \frac{k}{2v_0} \left[\mathfrak{b} + Bq_0 - \frac{h+\Phi_0}{\nabla_0} (\mathfrak{c} Ap_0 - \mathfrak{a} Cr_0) \right],$$

$$(\mathfrak{c}, \mathfrak{S}) = \frac{k}{2v_0} \left[\mathfrak{c} + Cr_0 - \frac{h+\Phi_0}{\nabla_0} (\mathfrak{a} Bq_0 - \mathfrak{b} Ap_0) \right].$$

Again,

$$\begin{aligned}(h, \mathfrak{S}) &= 2p(Ap, \mathfrak{S}) + 2q(Bq, \mathfrak{S}) + 2r(Cr, \mathfrak{S}). \\ (Ap, \mathfrak{S}) &= (Ap, \mathfrak{S}') + (Ap, \mathfrak{S}''), \\ (Ap, \mathfrak{S}') &= \frac{k}{\kappa v}(Ap, \kappa \varpi) + (Ap, k) \frac{d\mathfrak{S}'}{dk},\end{aligned}$$

therefore

$$\begin{aligned}(Ap, \kappa \varpi) &= \frac{1}{4} \{ (1 + \lambda^2)(\kappa u + 2\lambda \varpi) - (\lambda u + \varpi)\kappa \lambda \\ &\quad + (\lambda \mu + \nu)(\kappa v + 2\mu \varpi) - (\lambda v + w)\kappa \mu \\ &\quad + (\nu \lambda + \mu)(\kappa w + 2\nu \varpi) - (v + \lambda w)\kappa \nu \} \\ &= \frac{1}{4} \kappa(u + \lambda \varpi) = \frac{1}{2} \kappa(\mathfrak{a} + Ap),\end{aligned}\tag{55}$$

therefore

$$\begin{aligned}(Ap, \mathfrak{S}') &= \frac{k}{2v}(\mathfrak{a} + Ap), \\ (Ap, \mathfrak{S}'') &= -k \frac{h+\Phi}{v \nabla}(Ap, v) + \&c. \\ &= -\frac{1}{2} k \frac{h+\Phi}{v \nabla}(\mathfrak{b}Cr - \mathfrak{c}Bq) + Fv - Fv_0, \\ (Ap, \mathfrak{S}) &= \frac{k}{2v}[\mathfrak{a} + Ap - \frac{h+\Phi}{\nabla}(\mathfrak{b}Cr - \mathfrak{c}Bq)] + Fv - Fv_0,\end{aligned}\tag{56}$$

and similarly for $(Bq, \mathfrak{S})$, $(Cr, \mathfrak{S})$. Substituting, and neglecting the terms which vanish for $v = v_0$,

$$(h, \mathfrak{S}) = \frac{k}{v}(\Phi + h - \frac{\Phi+h}{\nabla} \nabla),$$

i.e.

$$(h, \mathfrak{S}) = 0.\tag{57}$$

Lastly, to find $(e, \mathfrak{S})$,

$$(e, \mathfrak{S}) = \{(e, \mathfrak{S})\} + (\mathfrak{a}, \mathfrak{S}) \frac{de}{d\mathfrak{a}} + (\mathfrak{b}, \mathfrak{S}) \frac{de}{d\mathfrak{b}} + (\mathfrak{c}, \mathfrak{S}) \frac{de}{d\mathfrak{c}},$$

where, in $\{(e, \mathfrak{S})\}$, the differentiations upon e are supposed not to affect the constants $\mathfrak{a}$, $\mathfrak{b}$, $\mathfrak{c}$. Neglecting the terms which vanish for $v = v_0$,

$$\begin{aligned}(e, \mathfrak{S}) &= \{(e, \mathfrak{S})\}, \\ \{(e, \mathfrak{S})\} &= \{(e, \mathfrak{S}')\} + \{(e, \mathfrak{S}'')\}, \\ \{(e, \mathfrak{S}')\} &= [\{(e, \mathfrak{S}')\}] + (e, k) \frac{d\mathfrak{S}'}{dk} = [\{(e, \mathfrak{S}')\}];\end{aligned}$$

where, in $[\{(e, \mathfrak{S}')\}]$, the differentiations upon e and $\mathfrak{S}$ do not affect the constants.

$$\{(e, \mathfrak{S}'')\} = [\{(e, \mathfrak{S}'')\}] + (e, \mathfrak{a}) \frac{d\mathfrak{S}''}{d\mathfrak{a}} + \&c.$$

neglecting the terms which vanish for $v = v_0$. therefore

$$\begin{aligned}(e, \mathfrak{S}) &= [\{(e, \mathfrak{S}')\}] + [\{(e, \mathfrak{S}'')\}] \\ &= [\{(e, \mathfrak{S}')\}];\end{aligned}$$

since

$$[\{(e, \mathfrak{S}'')\}] = (v, v) \frac{de}{dv} \frac{d\mathfrak{S}''}{dv} = 0.$$

Hence

$$(e, \mathfrak{S}) = -\frac{1}{2} \frac{k}{\kappa \nabla v} (v, \kappa \varpi), \quad (58)$$

$$\begin{aligned}(v, \kappa \varpi) &= \frac{1}{4} \{ [u + (1 + \kappa)\lambda\varpi](2\lambda u + \kappa u) - [\lambda\varpi^2 + (1 + \kappa)\varpi u]\kappa\lambda \\ &\quad + [v + (1 + \kappa)\mu\varpi](2\mu v + \kappa v) - [\mu\varpi^2 + (1 + \kappa)\varpi v]\kappa\mu \\ &\quad + [w + (1 + \kappa)\nu\varpi](2\nu w + \kappa w) - [\nu\varpi^2 + (1 + \kappa)\varpi w]\kappa\nu \} \\ &= \frac{1}{4} \{ 2\varpi^2 + \kappa(u^2 + v^2 + w^2) + 2(1 + \kappa)(\kappa - 1)\varpi^2 + \kappa(1 + \kappa)\varpi^2 \\ &\quad - \kappa(\kappa - 1)\varpi^2 - \kappa(\kappa + 1)\varpi^2 \} \\ &= \frac{1}{4} \kappa \{ (\kappa + 1)\varpi^2 + (u^2 + v^2 + w^2) \} = \frac{1}{4} 4\kappa v = \kappa v,\end{aligned} \quad (59)$$

therefore

$$(e, \mathfrak{S}) = -\frac{k}{\nabla}. \quad (60)$$

Hence, recapitulating,

$$\left. \begin{aligned}(\mathfrak{b}, \mathfrak{c}) &= -\mathfrak{a}, \quad (\mathfrak{c}, \mathfrak{a}) = -\mathfrak{b}, \quad (\mathfrak{a}, \mathfrak{b}) = -\mathfrak{c}, \\ (\mathfrak{a}, h) &= 0, \quad (\mathfrak{b}, h) = 0, \quad (\mathfrak{c}, h) = 0, \\ (\mathfrak{a}, e) &= -\frac{1}{\nabla_0} (\mathfrak{b}Cr_0 - \mathfrak{c}Bq_0), \\ (\mathfrak{b}, e) &= -\frac{1}{\nabla_0} (\mathfrak{c}Ap_0 - \mathfrak{a}Cr_0), \\ (\mathfrak{c}, e) &= -\frac{1}{\nabla_0} (\mathfrak{a}Bq_0 - \mathfrak{b}Ap_0), \\ (h, e) &= -2, \\ (\mathfrak{a}, \mathfrak{S}) &= \frac{k}{2v_0} \left[\mathfrak{a} + Ap_0 - \frac{h+\Phi_0}{\nabla_0} (\mathfrak{b}Cr_0 - \mathfrak{c}Bq_0) \right], \\ (\mathfrak{b}, \mathfrak{S}) &= \frac{k}{2v_0} \left[\mathfrak{b} + Bq_0 - \frac{h+\Phi_0}{\nabla_0} (\mathfrak{c}Ap_0 - \mathfrak{a}Cr_0) \right], \\ (\mathfrak{c}, \mathfrak{S}) &= \frac{k}{2v_0} \left[\mathfrak{c} + Cr_0 - \frac{h+\Phi_0}{\nabla_0} (\mathfrak{a}Bq_0 - \mathfrak{b}Ap_0) \right], \\ (h, \mathfrak{S}) &= 0, \\ (e, \mathfrak{S}) &= -\frac{k}{\nabla_0};\end{aligned} \right\} \quad (61)$$

and therefore

$$\left. \begin{aligned} \frac{d\mathfrak{a}}{dt} &= -\mathfrak{c} \frac{dV}{db} + \mathfrak{b} \frac{dV}{dc} - \frac{1}{\nabla_0} (\mathfrak{b}Cr_0 - \mathfrak{c}Bq_0) \frac{dV}{de} + \frac{k}{2v_0} \left[\mathfrak{a} + Ap_0 - \frac{h+\Phi_0}{\nabla_0} (\mathfrak{b}Cr_0 - \mathfrak{c}Bq_0) \right] \frac{dV}{d\mathfrak{S}}, \\ \frac{d\mathfrak{b}}{dt} &= -\mathfrak{a} \frac{dV}{dc} + \mathfrak{c} \frac{dV}{da} - \frac{1}{\nabla_0} (\mathfrak{c}Ap_0 - \mathfrak{a}Cr_0) \frac{dV}{de} + \frac{k}{2v_0} \left[\mathfrak{b} + Bq_0 - \frac{h+\Phi_0}{\nabla_0} (\mathfrak{c}Ap_0 - \mathfrak{a}Cr_0) \right] \frac{dV}{d\mathfrak{S}}, \\ \frac{d\mathfrak{c}}{dt} &= -\mathfrak{b} \frac{dV}{da} + \mathfrak{a} \frac{dV}{db} - \frac{1}{\nabla_0} (\mathfrak{a}Bq_0 - \mathfrak{b}Ap_0) \frac{dV}{de} + \frac{k}{2v_0} \left[\mathfrak{c} + Cr_0 - \frac{h+\Phi_0}{\nabla_0} (\mathfrak{a}Bq_0 - \mathfrak{b}Ap_0) \right] \frac{dV}{d\mathfrak{S}}, \\ \frac{dh}{dt} &= -2 \frac{dV}{de}, \\ \frac{de}{dt} &= \frac{1}{\nabla_0} \left[(\mathfrak{b}Cr_0 - \mathfrak{c}Bq_0) \frac{dV}{da} + (\mathfrak{c}Ap_0 - \mathfrak{a}Cr_0) \frac{dV}{db} + (\mathfrak{a}Bq_0 - \mathfrak{b}Ap_0) \frac{dV}{dc} \right] + 2 \frac{dV}{dh} - \frac{k}{\nabla_0} \frac{dV}{d\mathfrak{S}}, \\ \frac{d\mathfrak{S}}{dt} &= -\frac{k}{2v_0} \left[\left\{ \mathfrak{a} + Ap_0 - \frac{h+\Phi_0}{\nabla_0} (\mathfrak{b}Cr_0 - \mathfrak{c}Bq_0) \right\} \frac{dV}{da} \right. \\ &\quad + \left\{ \mathfrak{b} + Bq_0 - \frac{h+\Phi_0}{\nabla_0} (\mathfrak{c}Ap_0 - \mathfrak{a}Cr_0) \right\} \frac{dV}{db} \\ &\quad \left. + \left\{ \mathfrak{c} + Cr_0 - \frac{h+\Phi_0}{\nabla_0} (\mathfrak{a}Bq_0 - \mathfrak{b}Ap_0) \right\} \frac{dV}{dc} \right] + \frac{k}{\nabla_0} \frac{dV}{de}, \end{aligned} \right\} \quad (62)$$

to which we may join

$$\frac{dk}{dt} = \frac{dV}{d\mathfrak{S}}. \quad (63)$$

We have thus the complete system of formulae.

38.

NOTE ON A GEOMETRICAL THEOREM CONTAINED IN A PAPER BY SIR W. THOMSON.

[From the Cambridge and Dublin Mathematical Journal, vol. i. (1846), pp. 207, 208.]

It is easily shown that if three confocal surfaces of the second order pass through a point P , then the square of the distance of this point from the origin is equal to the sum of the squares of three of the axes, no two of which are parallel or belong to the same surface (the squares of one or two of the axes of the hyperboloids being considered negative); i.e. if

$$\frac{x^2}{a^2 + h} + \frac{y^2}{b^2 + h} + \frac{z^2}{c^2 + h} = 1,$$

$$\frac{x^2}{a^2 + k} + \frac{y^2}{b^2 + k} + \frac{z^2}{c^2 + k} = 1,$$

$$\frac{x^2}{a^2 + l} + \frac{y^2}{b^2 + l} + \frac{z^2}{c^2 + l} = 1;$$

then

$$x^2 + y^2 + z^2 = a^2 + b^2 + c^2 + h + k + l.$$

In fact these equations give

$$x^2 = \frac{(a^2 + h)(a^2 + k)(a^2 + l)}{(a^2 - b^2)(a^2 - c^2)},$$

$$y^2 = \frac{(b^2 + h)(b^2 + k)(b^2 + l)}{(b^2 - a^2)(b^2 - c^2)},$$

$$z^2 = \frac{(c^2 + h)(c^2 + k)(c^2 + l)}{(c^2 - a^2)(c^2 - b^2)};$$

and adding these and reducing, we have the relation in question; which is also immediately obtained by forming the cubic whose roots are h, k, l .